2F POST-TENSIONED SLAB - INDICATIVE DESIGN VERIFICATION (SCREENING)

Source design: Success Design & Build Co., Ltd - sheets 14-16 of 50, Rev 01, 22-Aug-2023 (2F PT layout and tendon profiles). Redrafted by CMED for review.

Item	Value
Report no.	TBC (project code not assigned in a design basis)
Revision / date	Draft A / 16-OCT-2026
Stage	INDICATIVE SCREENING - NOT A SIGN-OFF - NOT FOR CONSTRUCTION
Prepared by	Claude (AI-assisted calculation)
Checked by / Approved by	_____ (human engineer - not completed)
Design basis	No design-basis file exists for this project. Inputs are listed in sections C and D with their status (CONFIRMED / ASSUMED).

Status vocabulary: OK / NOT OK / NOT CHECKED (missing input). 'indicative' means the result depends on at least one ASSUMED input or a simplified method. Nothing in this report is APPROVED; a 'calculated OK' is not 'engineer approved'.

1. Answer in brief

Can this design be shown adequate? Not from the information available. Under the stated assumptions the screening finds no single failure that is certain, but it finds several items that do not pass and cannot be cleared without data that is not on the source sheets (mild reinforcement, stressing records, wall/core supports).

- Service stress (tension vs fctm): panel-average strip (case A) exceeds the limit at Y-direction midspan of the 9.5 m span (y 14.8 to 24.3 m) and, under the frequent combination, at the 10 m X-direction end spans. Utilisations are 1.0 to 1.1 - at the cracking threshold, not far beyond it. Column-strip bracket (case B) is worse (up to 1.7).
- Support (hogging) regions: with the 0.45 m drop-panel / band-beam section and the panel-average moment they pass (utilisation up to 0.4). With the Annex I column-strip share the C4 drop-panel supports exceed fctm (utilisation 1.2 to 1.3), and with the 0.25 m section alone they reach 1.4 to 1.6 (panel average) and up to 3.0 (column strip). The stress maps (separate PDF) show where.
- ULS flexure: the tendons alone cannot carry the ULS moments in several spans and supports. The design therefore relies on mild reinforcement that is not on the sheets supplied. Supplementary steel of about 0.1 to 2.3 x 10^3 mm2/m would be needed (section G4). NOT CHECKED, not 'failed'.
- Punching: 10 of 20 columns exceed vRd,c on tendon-only reinforcement. Interior C4 columns are marginal (1.02 to 1.10). Edge and corner C1 columns are 1.16 to 2.08 and depend strongly on how much slab they carry; wall/core support around the shafts is not modelled, so these are conservative. Mild steel or punching reinforcement is not given: NOT CHECKED.
- Deflection: uncracked strip-average deflection is 0.1 to 0.4 of L/250. OK (indicative); it would increase if the slab cracks.
- Load balancing: tendon profiles as drawn balance 98 % (X) and 86 % (Y) of self-weight - a reasonable range. Total drawn prestress is about 1.1 to 1.2 MPa on average, with local lows of 0.5 to 0.8 MPa.
- Drawing consistency: for 19 of 32 tendon marks the sheet-16 profile length does not equal the plan length (gap register G24). Until the engineer of record explains this, the profile-based load balancing is itself uncertain.
- Cracking: the service-stress result (at or just above the tensile strength at midspans) is consistent with cracks appearing under service load, and restrained shrinkage / temperature in a 47 m slab (section G8) can add up to about 3.3 MPa of net tension across the parameter range (G8). The screening cannot separate these causes. See section K.

A. Scope and structure

Level +3.100 slab, bonded PT flat slab 0.25 m thick, drop panels (DP) and band beams (BB1/BB2) 0.45 m thick, 47.0 m x ~21.2 m with 12 voids. Columns: C1 600, C3 800, C4 1100 mm (square assumed). Net slab area from the redrafted plan: 818.9 m2 (source states 829.05 m2; difference -1.2 %).

B. Codes

EN 1992-1-1 (CEN recommended values; no National Annex is recorded in a design basis), EN 1990 (combinations), EN 1991-1-1 (unit weight 25 kN/m3). Annex I of EN 1992-1-1 (informative) is used for column-strip / middle-strip shares. ACI / PTI are not used.

C. Materials and PT system

Symbol	Value	Unit	Source	Status
fck / fci	32 / 24	MPa	Drawing general note 1	CONFIRMED
Cover (to nearest reinforcement)	20	mm	Drawing note 1	CONFIRMED

Strand	12.7 mm, 7-wire, low relaxation, ASTM A416; Ap = 98.7	mm2	Drawing note 2 (grade not stated); skill reference value	CONFIRMED / grade ASSUMED
fpk (Gr.270)	1860	MPa	ASTM A416 Gr.270 typical (your choice)	ASSUMED - CONFIRM
fp0.1k	1674	MPa	0.9 fpk (skill: yield ~1675)	ASSUMED - CONFIRM
Jacking stress	0.75 fpk	-	Typical practice	ASSUMED - CONFIRM
Jacking force per strand	137.69	kN	calc	derived from assumed
Total losses	18 % (immediate 8 %)	-	Your choice (typical)	ASSUMED - CONFIRM
Effective force per strand Pe	112.90	kN	calc	derived from assumed
gamma_c / gamma_s	1.5 / 1.15	-	EN 1992-1-1 recommended	CEN recommended
Mild reinforcement	Not given (general note only: SD40 fy 400 MPa)	-	Drawing note 3	MISSING - quantities unknown
Exposure class	Not given; XC1 assumed for interior	-	-	MISSING
Creep coefficient phi	2.0	-	-	ASSUMED
Effective depth d (slab / thick zone)	0.20 / 0.40	m	t - 50 mm	ASSUMED

D. Actions

Action	Value (kN/m2)	Source
Self-weight slab 0.25 m x 25	6.25	EN 1991-1-1 Table A.1 (25 kN/m3); DP/BB extra weight not added (small, conservative omission)
SDL 250 KSM	2.45	Drawing plan note (1 kg/m2 = 0.00981 kN/m2)
LL 500 KSM	4.90	Drawing plan note. Note: PT-skill defaults are SDL 4.65 / LL 2.50 - they differ; sensitivity in section J

E. Combinations (EN 1990)

Combination	Expression	q (kN/m2)
ULS (6.10)	1.35 G + 1.5 Q	19.10
SLS frequent	G + psi1 Q (psi1 = 0.7, ASSUMED category C)	12.13
SLS quasi-permanent	G + psi2 Q (psi2 = 0.6, ASSUMED category C)	11.64
Transfer	Self-weight only, force at transfer	6.25

F. Method and provenance

- Geometry, tendon counts and lengths: transcribed from the source vector data (redraft_2f_pt.py, +/-0.3 m). Tendon profiles: sheet 16 (profiles_2F_PT.json, 683 stations).
- Prestress per section: strands crossing each section x Pe, divided by the net slab width at that section and 0.25 m (no credit for thicker zones).
- Load balancing: per tendon and span, w = 8 P a / L2 from the drawn profile; two drape definitions give a lower and an upper value (lower used). Each direction carries the full load (equivalent-frame practice); net load = combination minus balanced load.
- Strips: 1 m wide continuous strip on pinned supports (column stiffness neglected), Y strip supports at y = 4.3, 6.3, 14.8, 24.3 m; X strip at x = 0, 10, 19, 28, 37, 47 m. Solver verified against hand values (two equal spans: -wL2/8, 9wL2/128).
- Two cases bracket the answer: A = panel-average strip (moment on the full panel width); B = EN 1992-1-1 Annex I column strip (80 % hogging / 70 % sagging in a strip 0.25 x span each side), with the thick zone width 3.0-3.1 m at drop panels.
- Punching: reaction = q_ULS x tributary area, with the tributary area obtained by assigning every point of the net slab to the nearest column. Voids are excluded. Walls / cores are NOT modelled as supports (not identified on the sheets).
- ULS flexure: rectangular block, PT only, tendon stress fp0.1k/gamma_s; hyperstatic prestress moments NOT included.
- Calculation code: calc_lib/ec2_pt_screen.py and calc_lib/runs/pt_2f_screening.py, tests in tests/test_ec2_pt_screen.py (5 hand-value tests pass).

G. Checks

G1. Prestress and average precompression

Direction	Strands (total)	Strands per m (mean)	sigma_cp min / mean / max

			(MPa)
X	69	2.45	0.76 / 1.11 / 1.58
Y	136	2.58	0.51 / 1.16 / 2.13

Local lows: X at x = 5.8 (0.78), 6.2 (0.76), 12.8 (0.87), 13.2 (0.85), 34.2 (0.87), 40.2 (0.78), 40.8 (0.76), 41.2 (0.78) m; Y at y = 21.3 (0.84), 21.8 (0.82), 22.3 (0.84), 22.8 (0.84), 23.3 (0.83), 23.8 (0.83), 24.3 (0.51) m. EN 1992-1-1 does not set a minimum average prestress; these are reported as local lows only. X tendons are banded, so values between bands are lower and values in bands higher than this average.

G2. Load balancing

Direction	Balanced load (kPa, lower)	% of self-weight (lo-hi)	% of G (SW+SDL)	% of quasi-permanent
X	6.10	98 - 94 %	70 - 67 %	52 - 50 %
Y	5.39	86 - 83 %	62 - 60 %	46 - 45 %

The profile heights were scaled by plan length / profile length for the 19 marks where they disagree (G24); that scaling is a screening device, not a design value.

G3. Service stresses (tension against fctm = 3.02 MPa; fctk,0.05 = 2.12 MPa)

Tension shown, quasi-permanent / frequent. Utilisation = tension / fctm. Cracking can start below fctm; the 5 % fractile gives utilisations 1.43 times higher.

Dir	Location	Case	sigma_p MPa	m kNm/m (QP)	Tension MPa QP / FREQ	Util QP / FREQ	Status
Y	midspan 3 (8.5 m)	A avg-strip	1.30	17.4	0.37 / 0.50	0.12 / 0.17	OK (indicative)
Y	midspan 3 (8.5 m)	B col-strip	1.30	25.8	1.18 / 1.37	0.39 / 0.45	OK (indicative)
Y	midspan 4 (9.5 m)	A avg-strip	1.11	43.7	3.09 / 3.42	1.02 / 1.13	NOT OK (indicative)
Y	midspan 4 (9.5 m)	B col-strip	1.11	64.8	5.11 / 5.60	1.69 / 1.85	NOT OK (indicative)
Y	support 6.3 m	A avg-strip t=0.25	1.30	-22.6	0.87 / 1.04	0.29 / 0.34	OK (indicative)
Y	support 6.3 m	A avg-strip t=0.45	0.72	-22.6	0.00 / 0.00	0.00 / 0.00	OK (indicative)
Y	support 6.3 m	B col-strip, thick zone	0.72	-38.3	0.41 / 0.50	0.14 / 0.17	OK (indicative)
Y	support 6.3 m	B col-strip, t=0.25	1.30	-38.3	2.38 / 2.67	0.79 / 0.88	OK (indicative)
Y	support 14.8 m	A avg-strip t=0.25	1.09	-58.3	4.50 / 4.94	1.49 / 1.63	NOT OK (indicative)
Y	support 14.8 m	A avg-strip t=0.45	0.61	-58.3	1.12 / 1.25	0.37 / 0.41	OK (indicative)
Y	support 14.8 m	B col-strip, thick zone	0.61	-139.8	3.53 / 3.86	1.17 / 1.28	NOT OK (indicative)
Y	support 14.8 m	B col-strip, t=0.25	1.09	-98.7	8.38 / 9.12	2.77 / 3.02	NOT OK (indicative)
X	midspan 2 (10.0 m)	A avg-strip	1.43	44.1	2.80 / 3.17	0.92 / 1.05	NOT OK (indicative)
X	midspan 2 (10.0 m)	B col-strip	1.43	61.7	4.49 / 5.01	1.48 / 1.66	NOT OK (indicative)
X	midspan 3 (9.0 m)	A avg-strip	1.03	12.3	0.15 / 0.25	0.05 / 0.08	OK (indicative)
X	midspan 3 (9.0 m)	B col-strip	1.03	17.2	0.62 / 0.77	0.21 / 0.25	OK (indicative)
X	midspan 4 (9.0 m)	A avg-strip	0.97	22.2	1.16 / 1.35	0.38 / 0.45	OK (indicative)
X	midspan 4 (9.0 m)	B col-strip	0.97	31.1	2.01 / 2.28	0.67 / 0.75	OK (indicative)
X	midspan 5 (9.0 m)	A avg-strip	1.08	12.3	0.09 / 0.20	0.03 / 0.07	OK (indicative)
X	midspan 5 (9.0 m)	B col-strip	1.08	17.2	0.56 / 0.71	0.19 / 0.23	OK (indicative)
X	midspan 6 (10.0 m)	A avg-strip	1.43	44.1	2.80 / 3.17	0.92 / 1.05	NOT OK (indicative)
X	midspan 6 (10.0 m)	B col-strip	1.43	61.7	4.49 / 5.01	1.48 / 1.66	NOT OK (indicative)
X	support 10.0 m	A avg-strip t=0.25	1.01	-54.8	4.25 / 4.72	1.41 / 1.56	NOT OK (indicative)
X	support 10.0 m	A avg-strip t=0.45	0.56	-54.8	1.06 / 1.21	0.35 / 0.40	OK (indicative)
X	support 10.0 m	B col-strip, thick zone	0.56	-127.2	3.21 / 3.54	1.06 / 1.17	NOT OK (indicative)
X	support 10.0 m	B col-strip, t=0.25	1.01	-87.7	7.41 / 8.15	2.45 / 2.70	NOT OK (indicative)
X	support 19.0 m	A avg-strip t=0.25	0.98	-34.0	2.28 / 2.57	0.75 / 0.85	OK (indicative)
X	support 19.0 m	A avg-strip t=0.45	0.54	-34.0	0.46 / 0.55	0.15 / 0.18	OK (indicative)
X	support 19.0 m	B col-strip, thick	0.54	-78.9	1.79 / 2.00	0.59 / 0.66	OK (indicative)

		zone					
X	support 19.0 m	B col-strip, t=0.25	0.98	-54.4	4.24 / 4.70	1.40 / 1.55	NOT OK (indicative)
X	support 28.0 m	A avg-strip t=0.25	0.98	-34.0	2.28 / 2.57	0.75 / 0.85	OK (indicative)
X	support 28.0 m	A avg-strip t=0.45	0.54	-34.0	0.46 / 0.55	0.15 / 0.18	OK (indicative)
X	support 28.0 m	B col-strip, thick zone	0.54	-78.9	1.79 / 2.00	0.59 / 0.66	OK (indicative)
X	support 28.0 m	B col-strip, t=0.25	0.98	-54.4	4.24 / 4.70	1.40 / 1.55	NOT OK (indicative)
X	support 37.0 m	A avg-strip t=0.25	1.01	-54.8	4.25 / 4.72	1.41 / 1.56	NOT OK (indicative)
X	support 37.0 m	A avg-strip t=0.45	0.56	-54.8	1.06 / 1.21	0.35 / 0.40	OK (indicative)
X	support 37.0 m	B col-strip, thick zone	0.56	-127.2	3.21 / 3.54	1.06 / 1.17	NOT OK (indicative)
X	support 37.0 m	B col-strip, t=0.25	1.01	-87.7	7.41 / 8.15	2.45 / 2.70	NOT OK (indicative)

G4. ULS flexure (EN 1992-1-1 6.1), tendons only

MEd from the external ULS load on the strip; hyperstatic prestress moment not included; no mild steel credited. As,req is the supplementary bonded steel (fyd = 348 MPa, z = 0.9 d) needed to close the gap. Because mild reinforcement is not available, utilisations above 1.00 are NOT CHECKED, not failed.

Dir	Location	Case	MEd kNm/m	MRd (PT) kNm/m	Util	As,req mm2/m	Status
Y	support 6.3 m	A avg-strip	69.1	144.8	0.48	0	OK (indicative)
Y	support 6.3 m	B col-strip	117.1	144.8	0.81	0	OK (indicative)
Y	support 14.8 m	A avg-strip	178.0	144.8	1.23	265	NOT CHECKED (missing input)
Y	support 14.8 m	B col-strip	301.5	144.8	2.08	1251	NOT CHECKED (missing input)
Y	span 3 (8.5 m)	A avg-strip	53.3	70.8	0.75	0	OK (indicative)
Y	span 3 (8.5 m)	B col-strip	79.0	70.8	1.12	130	NOT CHECKED (missing input)
Y	span 4 (9.5 m)	A avg-strip	133.6	70.8	1.89	1003	NOT CHECKED (missing input)
Y	span 4 (9.5 m)	B col-strip	198.1	70.8	2.80	2033	NOT CHECKED (missing input)
X	support 10.0 m	A avg-strip	188.7	137.7	1.37	407	NOT CHECKED (missing input)
X	support 10.0 m	B col-strip	301.9	137.7	2.19	1311	NOT CHECKED (missing input)
X	support 19.0 m	A avg-strip	117.0	137.7	0.85	0	OK (indicative)
X	support 19.0 m	B col-strip	187.2	137.7	1.36	395	NOT CHECKED (missing input)
X	support 28.0 m	A avg-strip	117.0	137.7	0.85	0	OK (indicative)
X	support 28.0 m	B col-strip	187.2	137.7	1.36	395	NOT CHECKED (missing input)
X	support 37.0 m	A avg-strip	188.7	137.7	1.37	407	NOT CHECKED (missing input)
X	support 37.0 m	B col-strip	301.9	137.7	2.19	1311	NOT CHECKED (missing input)
X	span 2 (10.0 m)	A avg-strip	151.7	67.4	2.25	1346	NOT CHECKED (missing input)
X	span 2 (10.0 m)	B col-strip	212.4	67.4	3.15	2315	NOT CHECKED (missing input)
X	span 3 (9.0 m)	A avg-strip	42.2	67.4	0.63	0	OK (indicative)
X	span 3 (9.0 m)	B col-strip	59.1	67.4	0.88	0	OK (indicative)
X	span 4 (9.0 m)	A avg-strip	76.4	67.4	1.13	144	NOT CHECKED (missing input)
X	span 4 (9.0 m)	B col-strip	107.0	67.4	1.59	632	NOT CHECKED (missing input)
X	span 5 (9.0 m)	A avg-strip	42.2	67.4	0.63	0	OK (indicative)
X	span 5 (9.0 m)	B col-strip	59.1	67.4	0.88	0	OK (indicative)
X	span 6 (10.0 m)	A avg-strip	151.7	67.4	2.25	1346	NOT CHECKED (missing input)
X	span 6 (10.0 m)	B col-strip	212.4	67.4	3.15	2315	NOT CHECKED

							(missing input)

G5. Punching shear (EN 1992-1-1 6.4)

beta = 1.15 / 1.4 / 1.5 (interior / edge / corner, 6.4.3(6) recommended). vRd,c uses bonded PT only for rho_l (0.12 %) and k1 sigma_cp (sigma_cp = 1.13 MPa); no mild steel and no vertical tendon component credited. Square columns assumed. 'rho req' is the longitudinal reinforcement ratio that would make utilisation 1.00.

Column	Type	Trib m2	V kN	vEd MPa	vRd,c MPa	Util	rho req	Status
C1@(0,24.3)	corner	17.8	339	1.39	0.67	2.07	4.71 %	NOT CHECKED (missing input)
C1@(19,24.3)	edge	54.4	1039	0.84	0.56	1.52	1.41 %	NOT CHECKED (missing input)
C1@(28,24.3)	edge	55.5	1060	0.86	0.56	1.55	1.51 %	NOT CHECKED (missing input)
C1@(47,24.3)	corner	17.9	341	1.40	0.67	2.08	4.82 %	NOT CHECKED (missing input)
C1@(0,14.8)	edge	17.9	341	0.78	0.67	1.16	0.68 %	NOT CHECKED (missing input)
C1@(47,14.8)	edge	18.5	353	0.81	0.67	1.20	0.76 %	NOT CHECKED (missing input)
C1@(0,4.3)	corner	26.4	504	0.77	0.56	1.39	1.02 %	NOT CHECKED (missing input)
C1@(10,4.3)	edge	10.8	205	0.17	0.56	0.30	0.00 %	OK (indicative)
C1@(19,4.3)	edge	5.9	112	0.09	0.56	0.16	0.00 %	OK (indicative)
C1@(28,4.3)	edge	5.9	112	0.09	0.56	0.16	0.00 %	OK (indicative)
C1@(37,4.3)	edge	10.8	205	0.17	0.56	0.30	0.00 %	OK (indicative)
C1@(47,4.3)	corner	26.9	515	0.79	0.56	1.42	1.10 %	NOT CHECKED (missing input)
C3@(10,6.3)	interior	51.6	985	0.34	0.56	0.62	0.04 %	OK (indicative)
outside DP C3@(10,6.3)	interior		985	0.16	0.67	0.24	0.00 %	OK (indicative)
C3@(19,6.3)	interior	43.2	826	0.29	0.56	0.52	0.02 %	OK (indicative)
outside DP C3@(19,6.3)	interior		826	0.28	0.67	0.42	0.01 %	OK (indicative)
C3@(28,6.3)	interior	43.2	826	0.29	0.56	0.52	0.02 %	OK (indicative)
outside DP C3@(28,6.3)	interior		826	0.28	0.67	0.42	0.01 %	OK (indicative)
C3@(37,6.3)	interior	52.6	1005	0.35	0.56	0.63	0.05 %	OK (indicative)
outside DP C3@(37,6.3)	interior		1005	0.17	0.67	0.25	0.00 %	OK (indicative)
C4@(10,14.8)	interior	97.6	1865	0.57	0.56	1.02	0.34 %	NOT CHECKED (missing input)
outside DP C4@(10,14.8)	interior		1865	0.73	0.67	1.08	0.53 %	NOT CHECKED (missing input)
C4@(19,14.8)	interior	81.0	1547	0.47	0.56	0.85	0.17 %	OK (indicative)
outside DP C4@(19,14.8)	interior		1547	0.60	0.67	0.90	0.27 %	OK (indicative)
C4@(28,14.8)	interior	81.0	1547	0.47	0.56	0.85	0.17 %	OK (indicative)
outside DP C4@(28,14.8)	interior		1547	0.60	0.67	0.90	0.27 %	OK (indicative)
C4@(37,14.8)	interior	99.3	1897	0.58	0.56	1.04	0.37 %	NOT CHECKED (missing input)

outside DP C4@(37,14.8)	interior		1897	0.74	0.67	1.10	0.56 %	NOT CHECKED (missing input)

Face check vEd,u0 / vRd,max: largest utilisation 0.48 - OK (indicative).

G6. Deflection (EN 1992-1-1 7.4.1)

Span	Deflection mm	Direction	L/250 mm	Util	Status
Y-dir span 3 (8.5 m)	1.7	up	34.0	0.05	OK (indicative)
Y-dir span 4 (9.5 m)	13.5	down	38.0	0.36	OK (indicative)
X-dir span 2 (10.0 m)	15.8	down	40.0	0.40	OK (indicative)
X-dir span 3 (9.0 m)	4.9	up	36.0	0.13	OK (indicative)
X-dir span 4 (9.0 m)	2.4	down	36.0	0.07	OK (indicative)
X-dir span 5 (9.0 m)	4.9	up	36.0	0.13	OK (indicative)
X-dir span 6 (10.0 m)	15.8	down	40.0	0.40	OK (indicative)

Uncracked, Ec,eff = Ecm/(1+phi) = 11.1 GPa, strip-average, QP net load, column stiffness neglected. Not a cracked-section or two-way plate result; the L/500 after-construction limit (7.4.1(5)) cannot be checked without the partition / finish sequence.

G7. Transfer

Self-weight with the transfer force (Pt = 126.7 kN/strand, 8 % immediate loss ASSUMED, fci = 24 MPa): largest tension utilisation 0.00 (limit fctm(24) = 2.50 MPa) - OK (indicative). The net load is almost zero or upward in the X strip (net -0.59 kPa): the tendons nearly balance self-weight at transfer, so check the camber and the stressing sequence.

G8. Restraint, shrinkage and temperature (parametric - no code values)

A 47 m slab is tied to stiff cores / walls. No wall layout, pour sequence, shrinkage environment or temperature range is available, so strain values are PARAMETERS, not code values. sigma_t = R x eps x Ecm/(1+phi); net = sigma_t - mean X precompression. EN 1992-1-1 7.3.2 minimum reinforcement for restrained cracking (k = 1.0, kc = 1.0, fct,eff = 3.02 MPa, sigma_s = 400 MPa) is about 1.9 x 10^3 mm2/m (both faces) without prestress credit - cannot be compared as the mild steel is not given: NOT CHECKED.

Free strain (per mille)	Restraint R	Restraint tension (MPa)	Net after precompression (MPa)	Note
0.2	0.3	0.67	-0.44	residual compression
0.2	0.5	1.11	0.01	tension > 0
0.2	0.8	1.78	0.67	tension > 0
0.3	0.3	1.00	-0.10	residual compression
0.3	0.5	1.67	0.56	tension > 0
0.3	0.8	2.67	1.56	tension > 0
0.4	0.3	1.33	0.23	tension > 0
0.4	0.5	2.22	1.12	tension > 0
0.4	0.8	3.56	2.45	tension > 0
0.5	0.3	1.67	0.56	tension > 0
0.5	0.5	2.78	1.67	tension > 0
0.5	0.8	4.45	3.34	tension > 0

Correction 16-OCT-2026: an earlier draft applied the prestress force over the 0.25 m area also for the 0.45 m cases; P/A now scales with 1/t, which raises the thick-zone tensions slightly.

G9. Drawing consistency

Strand numbers 1 to 205 are continuous (69 X + 136 Y). Marks used on the plan all exist on sheet 16 (32 of 32). For 19 of 32 marks the profile length differs from plan label length minus 0.5 m by more than 0.3 m (gap register G24):

X2 (plan 35.55, profile 47.8 m); X3 (plan 48.4, profile 47.2 m); EX2 (plan 12.45, profile 12.1 m); X4 (plan 47.75, profile 10.52 m); X5 (plan 19.65, profile 11.2 m); X6 (plan 20.15, profile 10.52 m); EX3 (plan 14.4, profile 12.1 m); X7 (plan 19.65, profile 11.2 m); Y1 (plan 8.95, profile 21.2 m); Y1A (plan 8.75, profile 20.2 m); Y2 (plan 8.5, profile 20.2 m); Y3 (plan 8.45, profile 18.1 m); Y4 (plan 20.25, profile 15.6 m); Y5 (plan 17.6, profile 15.9 m); Y7 (plan 17.6, profile 19.2 m); Y8 (plan 17.9, profile 19.2 m); Y9 (plan 20.2, profile 18.2 m); Y9A (plan 20.7, profile 18.7 m); Y10 (plan 20.2, profile 18.7 m)

Cover: profile low points of 4 to 6 cm to the tendon centre allow 20 mm cover only if the duct is not deeper than about 40 mm - duct size TBC. High points of 25 to 27 cm occur in a 25 cm slab only inside the 0.45 m DP / BB; confirm on the redrafted plan.

H. Results summary

Check	Governing item	Max util	Status
Service tension - case A (panel average)	FREQ tension at support 14.8 m, Y-dir, A avg-strip t=0.25	1.63	NOT OK (indicative)
Service tension - case B (column strip)	FREQ tension at support 14.8 m, Y-dir, B col-strip, t=0.25	3.02	NOT OK (indicative)
ULS flexure - case A, PT only	ULS sag6, X-dir span 6 (10.0 m), A avg-strip, PT only	2.25	NOT CHECKED (missing input)

ULS flexure - case B, PT only	ULS sag6, X-dir span 6 (10.0 m), B col-strip, PT only	3.15	NOT CHECKED (missing input)
Deflection	Deflection X-dir span 6 (10.0 m), QP net, strip-average uncracked, Ec,	0.40	OK (indicative)
Punching u1	Punching u1, C1 600 mm at (47, 24.3), corner, slab, d=0.2	2.08	NOT CHECKED (missing input)
Punching u0 (vRd,max)	Punching u0 (vRd,max), C1 at (47, 24.3)	0.48	OK (indicative)

Overall: ENGINEER REVIEW REQUIRED. Not marked PASS: assumed PT inputs, missing mild reinforcement, simplified strip method.

I. Assumptions to confirm and limitations

- ASSUMED: fpk 1860 MPa, fp0.1k 1674 MPa, jacking 0.75 fpk, total loss 18 % (8 % immediate), creep phi 2.0, psi1 = 0.7 / psi2 = 0.6 (category C), exposure XC1, effective depths t - 50 mm, square columns, typical panel width 9 m. Each needs the engineer-of-record value.
- MISSING: stressing records and elongations, pour sequence and dates, mild reinforcement schedule, calculation / design report of the original designer, wall and core layout (support to the slab around the shafts), actual concrete strength test results, ground / environment data, loading from the partition layout.
- Method limits: pinned-support strips; no column or wall stiffness; hyperstatic moments omitted; no two-way plate (FE) analysis; column strip shares from an informative annex; thick zone modelled by thickness brackets; tendon profile effects only as load-balanced averages.
- The ADAPT / SAFE comparison required by the PT skill (section 34) has not been done; use this report only to choose what to check there.
- Sheet-16 profile lengths disagree with the plan for 19 marks (G24) - transcription error or design inconsistency not resolved.

J. Sensitivity to the assumed inputs

Variant	Pe kN/strand	q QP kPa	q bal X / Y	Service A >1	Service B >1	ULS A >1	Punching >1
base	112.9	11.64	6.1 / 5.4	10 / 38	22 / 38	7 / 13	10 / 20
skill-default loads (SDL 4.65 / LL 2.5)	112.9	12.40	6.1 / 5.4	12 / 38	22 / 38	7 / 13	9 / 20
losses 25%	103.3	11.64	5.6 / 4.9	12 / 38	23 / 38	7 / 13	10 / 20
losses 12%	121.2	11.64	6.5 / 5.8	7 / 38	20 / 38	7 / 13	10 / 20
d (punching) +10 mm	112.9	11.64	6.1 / 5.4	10 / 38	22 / 38	7 / 13	9 / 20

Counts are checks with utilisation above 1.00 out of the checks evaluated. Higher losses or higher loads add failing checks; lower losses remove a few but not the ULS or punching items.

Independent check record

- Pe = 0.75 x 1860 x 98.7 / 1000 x (1 - 0.18) = 112.9 kN - hand value matches.
- Y precompression at y = 8 m: 136 strands x 112.9 kN / (47.2 m x 0.25 m) = 1.30 MPa - matches the section table.
- C4 at (10, 14.8): V = 19.1 x 97.6 = 1864 kN; u1 = 4 x 1.1 + 2 pi x 0.8 = 9.43 m; vEd = 1.15 x 1864 / (9.43 x 0.40) = 0.57 MPa; vRd,c,min = 0.035 x 1.707^1.5 x 32^0.5 + 0.1 x 1.13 = 0.56 MPa - matches (util 1.02).
- Strip solver: two equal spans, support -wL2/8 and sagging 9wL2/128, and a 2 m cantilever -wc2/2 - all pass in tests/test_ec2_pt_screen.py.
- Not independently verified: balanced-load values (method UNVERIFIED against ADAPT), Annex I column strip shares, tributary allocation.

K. What this says about the cracks (not a conclusion)

The screening points to three mechanisms that can each produce the kind of cracking seen, and the available data cannot rank them:

- Service flexural tension at midspans and at supports where the effective thickness is only 0.25 m (utilisation about 1.0 to 1.5 on the panel-average strip). This is a design-stage cause (prestress level, tendon drape, section).
- Restrained shrinkage and temperature movement of a 47 m slab against cores / walls: 0 to 3.3 MPa net tension across the parameter range. PT shortening is also resisted by the same walls, which reduces the precompression that reaches the slab. This depends on the wall layout, pour strips and curing - mostly execution and detailing, partly design.

- Construction and stressing issues: low local precompression (0.5 to 0.8 MPa), stressing before the concrete reaches fci, tendon position errors, or the 19 profile / plan discrepancies if the tendons were built to the profile sheet. These need stressing records and as-built tendon levels.

To decide whether the root cause is the design, the next data needed are: the crack survey overlaid on the plan and these results, the stressing records, the reinforcement schedule and the original design calculation.